Assignment 4

Futures settlement and instantaneous-forward-rate yield curves

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1. Why OTC cash flows are discounted

The key is what the daily transfer represents. Conventional OTC collateral secures a future claim. Futures variation margin settles a daily gain or loss. A similar-looking cash transfer can therefore have different pricing consequences.

OTC: security for an unpaid claim

If an OTC trade is worth $10 to me and my counterparty posts $10 of cash collateral, the trade still has a $10 replacement value. The collateral must be returned or applied to the eventual payment, and I owe the poster the agreed collateral interest. It protects the claim; it is not an additional $10 of unconditional profit. Zero unsecured exposure does not mean zero trade value.

The future contractual payment still has time value. With ideal full cash collateralization in one currency, discounting uses the collateral remuneration rate c. For a terminal payment X_T,

$$V_t = E_t^{Q^c}[D_c(t,T)\, X_T], \quad D_c(t,T) = \exp[-\text{integral}_t^T c_s\, ds].$$

Here Q^c is the pricing measure consistent with those collateralized funding assumptions. If collateral interest is capitalized, a balance V_k becomes $(1 + c_k \text{ Delta } t)V_k$. The next top-up is $V_{k+1} - (1 + c_k \text{ Delta } t)V_k$. Paying the interest separately gives the same economics. This interest is why daily collateral transfers do not remove discounting.

Futures: settlement of realized profit or loss

If a long future rises from 100 to 110, I receive 10 as settlement of that gain. Immediately afterwards, the continuing contract has zero value. If I close at 110, I keep the settled profit; it is not returnable collateral backing an unpaid terminal claim. Initial margin is separate from this daily profit settlement.

Write $F_k = F(t_k,T)$. Under the assignment's assumption that the next period's discount factor D_k is known at t_k,

$$0 = E_k^Q[D_k(F_{k+1} - F_k)] = D_k(E_k^Q[F_{k+1}] - F_k).$$
$$F_k = E_k^Q[F_{k+1}] \quad \text{implies} \quad F(t_k,T) = E_k^Q[S_T].$$

Discounting was applied and then cancelled from a zero-value equation. This equation describes the quoted futures price. It does not say that future money has no time value. Rates still affect the underlying's carry and the investment or funding of daily settlements.

The cash-flow terms matter more than the venue. Some cleared OTC swaps settle to market but include price alignment interest or adjustments. The distinction above is for the conventional collateralized OTC trade and the standard future in the assignment. Sources [1]-[3] appear on page 3.

2. IFR model and discount factors

Construction. The value date is 2026-10-07. The SOFR component has ten annual IFR nodes; the funding component has a single zero-spread node and references SOFR. Tenors are adjusted Following on the US government-bond calendar. Curve times use ACT/ACT (ISDA). IFR inputs are annual decimals, with no further calibration or percentage conversion.

Left-continuous interpolation and integration

For increasing node times x_i, $f(x) = f_i$ on $(x_{i-1}, x_i]$. The first ordinate extends left and the last extends right. Interpolation uses the first knot greater than or equal to the query. The point gradient is one at that ordinate and zero elsewhere.

For integration, let w_i be the length of the part of $[0,T]$ using f_i. Then $\text{integral}_0^T f(s)\,ds = \text{sum}_i\, f_i w_i$, and its gradient is the vector w. Reversing the integration limits negates the weights; equal limits give zero.

$$D_{local}(T) = \exp[-\text{integral}_0^T f(s)\,ds]$$
$$D_{funding}(T) = D_{SOFR}(T)\, D_{funding,\,local}(T)$$

The model follows stored component references recursively and detects circular references. Zero funding IFR gives a local factor of 1, so the two complete curves agree. Flat extrapolation supplies the 15-year query beyond the last input node.

Component / tenor	Curve time (years)	IFR (decimal)
SOFR / 1Y	1.000000000	0.030
SOFR / 2Y	2.008840482	0.031
SOFR / 3Y	3.005479452	0.032
SOFR / 4Y	4.000000000	0.033
SOFR / 5Y	5.000000000	0.034
SOFR / 6Y	6.000643761	0.035
SOFR / 7Y	7.000000000	0.036
SOFR / 8Y	8.008219178	0.037
SOFR / 9Y	9.005479452	0.038
SOFR / 10Y	10.000643761	0.039
Funding spread / 1Y	1.000000000	0.000

Computed discount factors

Tenor / date	SOFR-1B	Funding local	SOFR-1B-FLAT
Value date / 2026-10-07	1.0000000000	1.0000000000	1.0000000000
6M / 2027-04-07	0.9851524245	1.0000000000	0.9851524245
1Y / 2027-10-07	0.9704455335	1.0000000000	0.9704455335
2Y / 2028-10-10	0.9405654378	1.0000000000	0.9405654378
5Y / 2031-10-07	0.8521559917	1.0000000000	0.8521559917
10Y / 2036-10-07	0.7082228717	1.0000000000	0.7082228717
15Y / 2041-10-07	0.5827649555	1.0000000000	0.5827649555

3. Cashflow price and analytic risk

The fixed-accrual product starts on 2026-10-07 and pays at its five-year termination date, 2031-10-07. Its notional is 100 and its accrual basis is ACT/360. The assigned payment is notional times accrual, with no additional coupon multiplier or principal repayment.

Quantity	Value
ACT/360 product accrual	5.072222222222
ACT/ACT (ISDA) curve time T	5.000000000000
Cashflow c = 100 x accrual	507.222222222222
Integrated SOFR IFR	0.159985680066
Complete funding discount factor	0.852155991697
Price V = c x discount factor	432.232455788284

Risk is calculated analytically, holding node times and c fixed. If w and u are the integral weights for the SOFR and funding inputs, the chain rule gives

$$V = c \exp[-\text{sum}_i f_i w_i - \text{sum}_j s_j u_j]$$
$$dV/df_i = -Vw_i, \quad dV/ds_j = -Vu_j.$$

Component	Tenor	IFR	Integral weight	dV/dX (unit decimal)
SOFR-1B	1Y	0.030	1.000000000	-432.23245579
SOFR-1B	2Y	0.031	1.008840482	-436.05359906
SOFR-1B	3Y	0.032	0.996638970	-430.77970953
SOFR-1B	4Y	0.033	0.994520548	-429.86405877
SOFR-1B	5Y	0.034	1.000000000	-432.23245579
SOFR-1B	6Y	0.035	0.000000000	0.00000000
SOFR-1B	7Y	0.036	0.000000000	0.00000000
SOFR-1B	8Y	0.037	0.000000000	0.00000000
SOFR-1B	9Y	0.038	0.000000000	0.00000000
SOFR-1B	10Y	0.039	0.000000000	0.00000000
SOFR-1B-FLAT	1Y	0.000	5.000000000	-2161.16227894

The single funding ordinate applies across all five curve years, so its derivative is -5V = -2161.16227894 despite its zero rate. The 6Y-10Y SOFR ordinates have zero exposure to this payment. Derivatives are per unit decimal IFR; the table has no 1 bp scaling.

Verification. All ten notebook code cells executed successfully. All nine independent tests passed, including hand-calculated interpolation weights, actual calendar dates, positive and negative funding spreads, reference recursion and cycles, 15-year extrapolation, invalid dates, and test-only central differences of all eleven risk parameters.

Scope. The starter reserves payment_currency for future use; it does not perform currency conversion. The notebook keeps the supplied USD inputs. With stochastic rates, a forward delivery price is E[DS]/E[D], while the assignment's futures quote is E[S] under its stated settlement assumptions; deterministic maturity discounting makes these prices agree.

References for the written answer

[1] BIS: settlement-to-market and collateralized-to-market derivatives.

[2] Lou: Derivatives Discounting Explained, Section 2.1.

[3] CME: Bank Asset/Liability Management: Swaps and Swap Futures, pp. 2-3.